ST ERIC HIGH SCHOOL

SERVICE AND EXCELLENCE | NORTON, ZIMBABWE

Comprehensive Online Platform Manual & Operational Blueprint

Technical Architecture, System Startup, Administration & Full Functional Reference

Document Purpose:	Official System Administration, Deployment & End-User Operations Manual
Target Audience:	Headmaster, Deputy Head, HODs, Teachers, Administrative Staff & IT Support
Platform Version:	Version 3.0 Enterprise Edition (Full ZIMSEC & Department Scoping Integration)
School Address:	Stand No 2395, Shonhiwa Drive, PBag 904, Norton, Zimbabwe (Tel: 062-2087 / 062-2200)
Authorisation:	Office of the Headmaster & Academic Board -- St Eric High School
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Executive Notice: *This platform unifies student admissions, department-level allocation, mark entry, ZIMSEC-compliant report cards, financial tracking, digital library, roll-call attendance, and multi-channel communications into a single, high-performance secure portal. All staff members must adhere to the data privacy and role scoping protocols detailed within this manual.*

Table of Contents & Executive Architecture

The St Eric High School Online Platform is an enterprise-grade School Management & Learning Platform custom-engineered for the secondary education environment in Zimbabwe. The system directly implements ZIMSEC (Zimbabwe School Examinations Council) curriculum specifications, dual-tier academic streams (O-Level Forms 1-4 and A-Level Forms 5-6), department-isolated Head of Department (HOD) governance, and high-fidelity terminal report card production.

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Core Design Philosophy & Institutional Branding

Following administrative directives, the user interface rejects generic software aesthetics in favour of a prestigious academic design language inspired by premier regional institutions such as the University of Zimbabwe:

- [-] **Authentic School Identity:** Primary palette established around Navy Blue (#1a3563) and Crimson Red (#c41e2a), accented with subtle academic gold and crisp neutral paper backgrounds.
- [-] **Zero Casual Emoticons:** Strict eradication of informal emojis; replaced entirely by refined, high-precision Lucide monochrome SVG vector iconography.
- [-] **Department Governance:** Heads of Department (HODs) have dedicated authority to manage teacher allocations and curricula within their specific faculty without administrative interference.
- [-] **Zimbabwe Examination Alignment:** Strict compliance with official ZIMSEC syllabus subject naming, code numbers, and distinctive O-Level vs. A-Level grading bands.

Section 1: System Architecture & Technical Stack

The St Eric High School Online Platform utilizes a high-efficiency decoupled architecture designed for rapid local network operation or high-availability cloud deployment. The system combines an asynchronous Python backend with a reactive TypeScript frontend.

Component	Technology	Architectural Responsibility
API Server	FastAPI (Python 3.11/3.14)	REST API routing, request validation via Pydantic v2, GZip compression, token verification.
ORM & Database	SQLAlchemy 2.0 + SQLite / PostgreSQL	Data persistence, UUID primary keys, connection pooling, optimistic version concurrency.
Security & Auth	python-jose + passlib (bcrypt)	HS256 JWT bearer tokens containing subject ID, role claims, and 480-minute validity.
Frontend Web	React 18 + TypeScript 5	Component tree, single-page application (SPA) routing with react-router-dom v6.
Styling Engine	Tailwind CSS + Custom Design System	Utility-first responsive styles, branded gradients, custom scrollbar styling, print engine.
Build Tooling	Vite 5	Hot Module Replacement (HMR), production bundle minification (398 kB JS, 52 kB CSS).
Document Engine	ReportLab + PDF Generator	Terminal student report card generation, official transcripts, and administrative manuals.

Security Architecture & Request Lifecycle

Every incoming HTTP request undergoes a multi-stage security pipeline before reaching business logic:

1. **Context Middleware:** Generates a unique X-Request-ID header and enforces security headers (X-Content-Type-Options: nosniff, Referrer-Policy: no-referrer).
2. **CORS Filter:** Restricts cross-origin interactions exclusively to authorized domain origins configured in settings.CORS_ORIGINS.
3. **JWT Authentication (current_user):** Validates bearer token signature and expiration, ensuring user status is active.
4. **Role & Department Enforcement:** Validates whether the authenticated user possesses clearance for the endpoint and checks departmental boundaries.
5. **Audit Event Logging:** Records modifications (mark alterations, user role updates, student promotions) in the AuditEvent and MarkAudit tables.

Filesystem Organization

[-] backend/app/models.py: Unified relational database models.

[-] backend/app/routers/: Granular API routers (auth, setup, marks, people, misc, enrolment, etc.).

[-] frontend/src/pages/: Application route views (Dashboard, MarkEntry, ReportCards, Setup, Apply, etc.).

[-] frontend/src/components/: Reusable UI widgets (Shell, ui.tsx, PageHeader, Stat, Badge).

[-] scripts/: Standalone database seeding, migration, and maintenance utilities.

Section 2: Installation & Startup Guide

Follow this exact step-by-step procedure to launch the system on any Windows, macOS, or Linux machine:

Step 1: System Prerequisites

[-] **Python 3.10+** (with pip) and **Node.js 18+** (with npm) must be installed and on PATH.

Step 2: Backend Dependencies & Environment Setup

Open terminal (PowerShell or Bash) and navigate to the backend directory:

```
cd "c:\Users\Mr Ex\Desktop\St Eric High Platform\ST ERIC ONLINE PLATFORM FINAL\backend"
pip install -r requirements.txt
```

Verify or create backend/.env with standard configuration values:

```
SECRET_KEY=st-eric-high-school-production-secure-key-2026
DATABASE_URL=sqlite:///./dev.db
ACCESS_TOKEN_MINUTES=480
CORS_ORIGINS=http://localhost:5173,http://127.0.0.1:5173
UPLOAD_DIR=./uploads
```

Step 3: Database Seeding & Initialization

Populate the database with all 45 ZIMSEC subjects, 5 HODs, 8 teachers, and 27 active students:

```
python ..\scripts\seed_hod_and_subjects.py
python ..\scripts\seed_demo_marks.py
```

Step 4: Launching the Backend API Server

Start Uvicorn with auto-reload enabled (runs on port 8000):

```
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

API Verification: Open http://127.0.0.1:8000/docs to view interactive Swagger documentation.

Step 5: Frontend Dependencies & Dev Server Launch

Open a second terminal window and navigate to the frontend directory:

```
cd "c:\Users\Mr Ex\Desktop\St Eric High Platform\ST ERIC ONLINE PLATFORM FINAL\frontend"
npm install
npm run dev -- --host --port 5173
```

Access Portal: Open http://localhost:5173 in Google Chrome, Microsoft Edge, or Firefox.

Step 6: Production Build Verification

Run npm run build in the frontend folder to compile TypeScript without errors and produce minified assets in frontend/dist/.

Section 3: User Roles & Access Hierarchy

The platform enforces strict Role-Based Access Control (RBAC). A user's granted role determines sidebar navigation items, page accessibility, and backend database permissions.

Role	Key Responsibilities	Permitted Modules & Actions
Headmaster / Admin (admin)	Supreme institutional oversight, policy enforcement, student promotion.	[-] Full access to all 16 platform modules. [-] School-wide mark overriding & audit inspection. [-] Report card publication authorization. [-] Annual grade promotion & bulk import. [-] Admissions review & acceptance.
Head of Department (hod)	Faculty administration, teacher class allocation, departmental monitoring.	[-] Department Dashboard (teachers, subjects, classes count). [-] Assigns faculty teachers to classes. [-] Scoped view: Sciences, Commercials, Humanities, Languages, or Voc Tech. [-] Learner performance analytics for department.
Subject Teacher (teacher)	Class instruction, continuous assessment, attendance roll calls.	[-] Mark entry for assigned classes and subjects. [-] Daily class attendance roll call recording. [-] Homework creation and submission grading. [-] AI-assisted learner comments generation.
Librarian (librarian)	Digital syllabus archive, past papers, curriculum materials curation.	[-] Resource upload, categorization, moderation. [-] Review pending teacher uploads. [-] Digital library catalogue management.
Student (student)	Academic learning, homework submission, personal results tracking.	[-] Personal dashboard & class timetable. [-] Published terminal report cards download. [-] Homework submission portal. [-] Digital library resource access. [-] Fee balance overview.
Parent / Guardian (parent)	Guardian oversight, fee compliance, attendance tracking.	[-] Multi-child dashboard overview. [-] Real-time attendance rate tracking. [-] Official report cards viewing & PDF download. [-] Fee balance ledger & receipt statements. [-] School consent notices signing.

Department Boundary Scoping Principle

A critical innovation in Version 3.0 is **Department Boundary Isolation**. To maintain institutional discipline:

[-] An HOD can only view teachers and subjects tagged with their specific faculty.

[-] When creating teaching assignments, the backend strictly verifies that both the selected teacher and the selected subject match the HOD's department.

[-] Any attempt by an HOD to assign a teacher or subject outside their jurisdiction is immediately intercepted and blocked with a 403 Forbidden security exception.

Section 4: Every Function Fully Explained

4.1 Admissions & Enrolment Portal (/apply & /applications)

The Admissions subsystem streamlines prospective learner intake from public submission through to registered student status:

[-] **Public Application Interface (/apply):** Open to parents and guardians without requiring pre-existing login credentials.

[-] **Comprehensive Form Support:** Allows applications across all six secondary forms (Form 1 through Form 6).

[-] **Form 5 Advanced Level Rule:** The 'Subjects of Interest' field is dynamically enabled *only* when Form 5 is selected, allowing A-Level candidates to state their intended subject combination (Sciences, Commercials, or Arts). For Forms 1-4, this field is intelligently disabled.

[-] **Mandatory Document Verification:** Applications cannot be submitted without uploading both the learner's official Birth Certificate and the Last School Term Report Card. Frontend validation blocks empty uploads, and the backend returns a 422 Unprocessable Entity if either file is omitted.

[-] **Live Application Tracking:** Upon submission, applicants receive a unique tracking reference (e.g. SE-2026-X8K9L2). Guardians can check real-time application status at any time using their reference code and phone number.

[-] **Headmaster Review Queue (/applications):** Administrators review submitted applications, inspect uploaded PDF/image credentials, and either Accept or Decline. Upon acceptance, the system automatically creates a new Student profile, a linked User account, and a Parent account with default credentials.

4.2 Academic Setup & Roster Management (/setup)

The Setup module configures the foundational architecture of the school year:

[-] **Stream-Based Class Architecture:** Forms 1 to 4 utilize streams A, B, C, D, E, F, G, H (e.g., 1A, 1B, 2A, 3C, 4A). Forms 5 and 6 utilize academic faculty streams: Sciences A, Sciences B, Commercials A, Commercials B, and Arts.

[-] **Class Initialization:** A single click of 'Initialize Standard Classes' automatically establishes all 42 standard forms and streams for the academic year.

[-] **Bulk Roster Import:** Administrative staff can upload multi-column CSV or Excel spreadsheets containing hundreds of student or staff records for instant provisioning.

[-] **Annual Promotion Engine:** Advances students from Form 1 to Form 2, Form 2 to Form 3, etc., while transitioning Form 4 graduates into Form 5 screening and archiving Form 6 graduates.

4.3 ZIMSEC Curriculum & Subject Catalogue

The curriculum database includes 45 official ZIMSEC subjects with official national syllabus code numbers:

Department	Official ZIMSEC Subjects Seeded in Platform
Sciences	Mathematics (4004), Additional Mathematics (4034), Combined Science (4003), Physics (5054), Chemistry (5070), Biology (5090), Computer Science (7014), Pure Mathematics (A-Level), Statistics (A-Level), Physics (A-Level), Chemistry (A-Level), Biology (A-Level), Software Engineering.
Commercials	Commerce (7103), Principles of Accounting (7112), Economics (2281), Business Enterprise Skills (4048), Accounting (A-Level), Economics (A-Level), Management of Business (A-Level).
Humanities	Geography (2248), History (2167), Family & Religious Studies (2046), Heritage Studies (4006), Literature in English (2013), Divinity (A-Level), History (A-Level), Geography (A-Level), Literature (A-Level).
Languages	English Language (4005), Shona (3159), Ndebele (3155), Shona Language & Literature (A-Level).
Practical Arts	Agriculture (5038), Building Technology & Design (6043), Technical Graphics & Design (6042), Textile Technology & Design (6050), Food Technology & Design (6064), Metal Technology & Design (6040), Wood Technology & Design (6045), Music (6020), Art & Design (6015), Physical Education (6080), Crop Science (A-Level), Animal Science (A-Level).

4.4 Head of Department (HOD) Operations

Each academic department has an autonomous Head of Department login (e.g. hod.sciences, hod.commercials):

[-] **Departmental Dashboard:** Displays live metrics on active teachers in the department, total curriculum subjects, and active class allocations.

[-] **Teacher Class Allocation:** The HOD selects a teacher within their department, a subject belonging to their curriculum, and a target class stream. Clicking 'Assign Teacher' creates the official teaching assignment.

[-] **Department Filter Toolbar:** In the Curriculum tab, HODs can instantly filter syllabus subjects by their department or view the complete school offerings.

4.5 Assessment, Mark Entry & ZIMSEC Grading Scale (/marks)

The mark entry engine accurately implements the dual grading scales utilized across Zimbabwe:

Level	Grade	Percentage Band	ZIMSEC Descriptor & Standing
O-Level (Forms 1-4)	**A**	75% - 100%	Distinction (Highest Academic Achievement)
O-Level	**B**	60% - 74%	Credit (Meritorious Performance)
O-Level	**C**	50% - 59%	Pass (Official ZIMSEC Subject Pass Mark)
O-Level	**D**	45% - 49%	Marginal Fail (Near Pass Standard)
O-Level	**E**	40% - 44%	Weak Performance (Substantial Remedy Needed)
O-Level	**U**	0% - 39%	Ungraded (Unclassified Fail)
A-Level (Forms 5-6)	**A**	75% - 100%	Distinction (5 Points)
A-Level	**B**	65% - 74%	Very Good (4 Points)
A-Level	**C**	55% - 64%	Good (3 Points)
A-Level	**D**	45% - 54%	Satisfactory (2 Points)
A-Level	**E**	40% - 44%	Subsidiary Pass (1 Point)
A-Level	**O**	35% - 39%	Ordinary Level Pass Equivalence (0 Points)
A-Level	**F**	0% - 34%	Fail (Unclassified)

Mark Entry Operations & Immutable Audit Trail

[-] Teachers access /marks, filter by their assigned Class and Subject, and choose the Assessment (Assignment, Test, Mid-Term, or Exam).

[-] Entering a numeric mark triggers instant client-side and server-side grade calculation according to the class form level.

[-] Any mark alteration is recorded in MarkAudit with previous value, new value, actor user ID, and timestamp, guaranteeing complete institutional integrity.

4.6 Official Report Card Engine (/report-cards)

[-] **Physical Report Replica:** Styled directly against the St Eric High School physical report card format, complete with school crest, student personal metadata, attendance summary, subject marks, grades, and teacher signatures.

[-] **Dual Teacher & Headmaster Remarks:** Class teachers and the Headmaster enter qualitative remarks evaluating student conduct and progress.

[-] **AI-Assisted Comment Drafting:** Subject and class teachers can leverage Anthropic Claude-powered comment generation to draft constructive, culturally tailored remarks based on the learner's actual grade trajectory.

[-] **Publication Governance:** Report cards remain in 'Draft' status until formally released by the Headmaster. Once published, students and parents can download high-resolution printable PDFs from their dashboard.

4.7 Learner Analytics & Best Performer per Form (/insights)

[-] **Best Performing Student per Form:** The dashboard and insights module automatically aggregates marks and computes weighted GPA across all subjects to identify and display the top-ranked scholar in Form 1, Form 2, Form 3, Form 4, Form 5, and Form 6.

[-] **Class Distribution Curves:** Visualizes mark spreads, standard deviations, and subject pass rates.

[-] **Academic Intervention Alerts:** Automatically flags students exhibiting sharp performance declines across consecutive terms.

4.8 Daily Attendance Tracking (/attendance)

[-] **One-Click Roll Call:** Class teachers select their class and date to view an alphabetized student register. One click toggles status between Present, Absent, Late, and Excused.

[-] **Cumulative Attendance Percentage:** Computes exact attendance percentages per student for inclusion on official terminal report cards.

[-] **Parent Notification:** Unexcused absences can trigger automated notification alerts to registered guardians.

4.9 Digital Homework & Online Submissions (/assignments)

[-] **Assignment Creation:** Teachers create tasks with instructions, due dates, maximum marks, and downloadable attachments.

[-] **Learner Submissions:** Students upload electronic work before the deadline, viewing their submission status and timestamp.

[-] **Teacher Marking:** Teachers assess submissions online, assign marks, and provide written feedback.

4.10 Digital Library & Syllabus Repository (/library)

[-] **Resource Categories:** Past ZIMSEC examination papers, marking schemes, curriculum syllabi, and supplementary textbooks.

[-] **Librarian Moderation Queue:** Teacher-uploaded materials enter a pending moderation state until reviewed and approved by the Librarian.

[-] **Student Discovery:** Indexed search by subject, form level, and resource type.

4.11 Finance & Fee Ledger Management (/fees)

[-] **Term Balance Statements:** Displays tuition fees billed, levies, payments received, and outstanding balance.

[-] **Payment Logging:** Administrative staff record bank transfers, cash payments, or Ecocash reference numbers.

[-] **Parent Ledger Visibility:** Parents can view an itemized statement of payments and download official receipts.

4.12 School Communications & Consent Notices (/notices & /announcements)

[-] **Broadcast Announcements:** Headmaster notices broadcast instantly to students, teachers, or parents.

[-] **Consent Notices:** Official notices requiring electronic sign-off from parents (e.g. school excursions, medical consent, sporting tours).

[-] **Real-Time Bell Notifications:** In-app alerts notify users of new marks, homework assignments, or urgent announcements.

4.13 ZIMSEC Candidate Management (/candidates)

[-] **Candidate Profiles:** Records candidate number, national ID, birth entry number, and examination centre number.

[-] **Subject Entry Registration:** Enrolls candidates into specific ZIMSEC examination papers and tracks exam timetables.

4.14 Master Timetable System (/timetable)

[-] **Class Timetable:** Visual weekly grid (Periods 1 through 8, Monday to Friday) for every class stream.

[-] **Teacher Timetable:** Shows individual teaching rosters to prevent scheduling clashes.

Section 5: Default Credentials & Troubleshooting

The following master table lists the pre-configured system accounts established during database initialization. All accounts use the default password admin123 (users are prompted to change their password on first login).

Username	Full Name / Title	Role	Department / Jurisdiction	Password
admin	System Administrator (Headmaster)	admin	All Institutional Operations	admin123
hod.sciences	HOD Sciences	hod	Sciences (Maths, Bio, Chem, Phys, CS)	admin123
hod.commercials	HOD Commercials	hod	Commercials (Accounts, Commerce, Econ)	admin123
hod.humanities	HOD Humanities	hod	Humanities (Geog, Hist, FRS, Heritage)	admin123
hod.languages	HOD Languages	hod	Languages (English, Shona, Ndebele)	admin123
hod.voctech	HOD Practical Arts	hod	Practical Arts (Agric, Building, Graphics)	admin123
teacher1	Dr. C. Shumba	teacher	Sciences (Senior Physics & Maths)	admin123
teacher2	Mrs M. Sibanda	teacher	Languages (English Language)	admin123
teacher3	Mr K. Ndlovu	teacher	Humanities (History & Heritage)	admin123
teacher4	Mrs T. Chidzero	teacher	Commercials (Accounting & Commerce)	admin123
librarian	Mrs R. Musonza	librarian	School Digital Library	admin123
st.001	Takudzwa Sibanda	student	Form 1 Stream A (Student)	admin123
parent1	Mr. T. Sibanda (Guardian)	parent	Guardian to Takudzwa Sibanda	admin123

Troubleshooting & Operational FAQ

Q1: Backend displays WinError 10048: Only one usage of each socket address

Resolution: An orphaned Python or Uvicorn process is already holding port 8000. In PowerShell, run Get-Process -Name python* | Stop-Process -Force or identify the PID with Get-NetTCPConnection -LocalPort 8000 and terminate it.

Q2: How do I backup the entire database and all uploaded documents?

Resolution: SQLite stores all records in a single file: backend/dev.db. Student documents are in backend/uploads/. Create an immediate backup using: copy backend\dev.db backend\dev.db.backup.

Q3: What should an administrator do if a teacher forgets their password?

Resolution: Log into the Headmaster account (admin), navigate to **Setup -> System Users**, locate the user in the table, and click **Reset Password**. The user's password resets to admin123 with must_change_password=True.

Certification of Operability: *This document certifies that the St Eric High School Online Platform Version 3.0 has undergone full integration testing, database seeding, ZIMSEC scale verification, and role boundary validation. The system is certified ready for institutional operations.*

Signed: Academic Board & Office of the Headmaster -- St Eric High School, Norton